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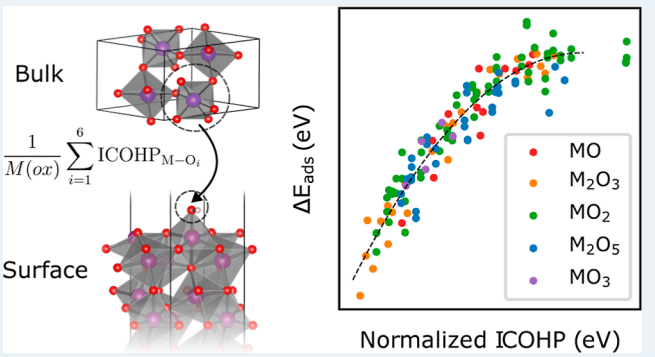
Prediction of O and OH Adsorption on Transition Metal Oxide Surfaces from Bulk Descriptors

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ABSTRACT: In the search for stable and active catalysts, density functional theory and machine learning-based models can accelerate the screening of materials. While stability is conveniently addressed on the bulk level of computation, the modeling of catalytic activity requires expensive surface simulations. In this work, we develop models for the surface adsorption energy of O and OH intermediates across a consistent and extensive data set of pure transition metal oxide surfaces. We show that adsorption energies across metal oxidation states of +2 to +6 are well captured from the metal−oxygen bond strength extracted from the bulk level calculation. Specifically, we calculate the integrated crystal orbital Hamiltonian population (ICOHP) of the metal−oxygen bond in the bulk oxide and employ a simple normalization scheme to obtain a strong correlation with the adsorption energetics. By combining our ICOHP descriptor with non-DFT features in a Gaussian Process regression (GPR) model, we achieve a high model accuracy with mean absolute errors of 0.166 and 0.219 eV for OH and O adsorption, respectively. By targeting the adsorption energy difference of the OH−OH adsorption with our GPR model, we predict the oxygen evolution reaction activity from bulk descriptors only. Furthermore, we utilize the strong correlation between the COHP and metal−oxygen bond lengths to rapidly predict the adsorption energetics and catalytic activity from the optimized bulk geometry. Our approach can enable an efficient search for active catalysts by eliminating the need for surface calculations in the initial screening phase.



KEYWORDS: *adsorption energy descriptors, machine learning, oxygen evolution reaction, transition metal oxides, density functional theory, crystal orbital Hamiltonian population*

The discovery of high-performing catalysts for the oxygen reduction and oxygen evolution reactions (ORRs/OERs) and related reactions involving oxygen is critical for the development of energy-efficient hydrogen fuel cells and electrolyzers.[1−4] Most efficient catalysts generally consist of precious metals (Pt) and precious metal oxides (IrOx, RuOx) with near-optimal adsorption energetics resulting in high Faradaic efficiencies and currents. Clearly, more abundant and affordable alternatives must be discovered to advance large-scale and cost-competitive hydrogen technologies. In the search for new stable and active catalysts, computational materials and catalysis simulations using density functional theory (DFT) play important roles in the screening of a larger material space at minimal cost. In these approaches, the adsorption energies of O and OH are the most important DFT-based descriptors for the ORR and OER catalytic activity.[5,6] Adsorption energies in general require performing surface DFT calculations with order of magnitude higher cost than bulk calculations, especially when considering the large structural and compositional variations of transition metal compounds.[7] Therefore, the development of surrogate models for adsorption energetics can help to further accelerate the exploration of catalytic materials by eliminating the need for surface DFT calculations in the initial screening phase. To achieve this goal, the generation of high-quality data sets and the development of open databases for catalysis also play a crucial role.[8−10]

Ideally, surrogate models for surface reactivity should be accurate, transferable, and interpretable, thus allowing for the development of design principles to help guide the discovery of new catalysts. In this regard, the identification of relevant material descriptors, particularly those based on electronic structure, plays an important role in this field. In the case of O

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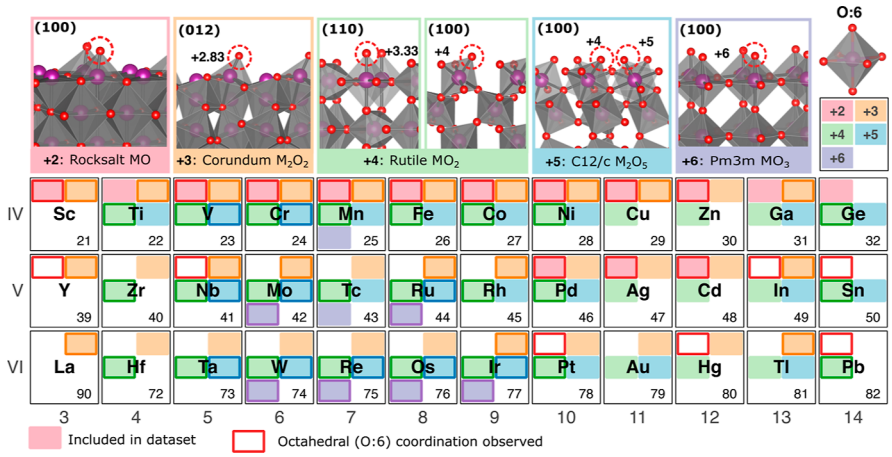


Figure 1. Overview of metal oxides and their surfaces used for the calculation of *O and *OH adsorption energy data set. Selected metal oxides and their surfaces (fully shaded) span 3d, 4d, 5d transition metals and two columns of p-block post-transition metal elements and range in oxidation states from +2 to +6. The most common crystal structure polymorphs with octahedral M−O coordination were chosen for each oxidation state for a given section of the periodic table. Our selection is justified by the experimental observation of a dominant octahedral coordination[11] highlighted as colored rectangle outlines.

and OH adsorption of transition metal oxides, several electronic-structure based descriptors have been discussed in the literature including: eg-orbital filling,[12] metal d-band center and widths,[13] the oxygen 2p-band center,[14] as well as combinations thereof.[15] Recent works have shown that the integrated crystal orbital Hamiltonian populations (ICOHP)[16,17] of the metal−oxygen bonds at the surface trend well with the adsorption energies on surfaces.[18] Furthermore, we recently found that the ICOHP drawn from bulk materials is a good descriptor for surface adsorption energies that qualitatively and quantitatively capture the trends in O and OH adsorptions.[19] For this reason, ICOHP has the potential to provide an estimate of the OER and ORR activity for a given catalyst from bulk calculations, thus accelerating the data-driven exploration of materials for water and oxygen catalysis. However, the generalizability of these correlations across oxidation states, bulk polymorphs, surface facets, and bonding types has yet to be investigated.

In this work, we extend our study to pure oxides of multiple oxidation states (+2, +3, +4, +5, and +6) to demonstrate the ability of ICOHP to predict surface adsorption energies of O and OH, through the generation of an extensive data set for adsorption energies. We show how a new normalization scheme for the bulk ICOHP enables comparison across different oxidation states and that integration with bulk and non-DFT surface descriptors allows for generalized predictions across different catalytic surfaces. We additionally discover a direct analytical relationship between the metal−oxygen bond length and ICOHP that is found to be universal for a given metal. Finally, we combine these findings to generate a machine learning (ML) model that can estimate surface adsorption energies and OER activity using only the bond length of the bulk structures. Thus, our main findings are that the bulk M−O ICOHP is highly correlated to surface adsorption and that high precision models can be achieved when combining the ICOHP with bulk and non-DFT surface descriptors in a surrogate model. This work shows a path forward toward predicting a material's OER/ORR activity based primarily on bulk calculations, avoiding the complexity and cost of surface calculations in the early material screening phases.

RESULTS AND DISCUSSION

Adsorption Energy Data Set. We have generated an extensive data set for the adsorption energies of O and OH across pure transition metal and late metal oxides. The data set spans oxidation states from +2 to +6, choosing the most frequently occurring crystal structure polymorph for each oxidation state as summarized in Figure 1. We note that while the entire transition metal series has been studied in +3, +4 and +5 oxidation states, a smaller subset was chosen for +2 and +6, due to the limited prevalence of these oxidation states. The +4 (rutile) calculations have been adopted from our previous work.[19]

We selected stable (low surface-energy) facets of rocksalt, corundum, rutile, and highly oxidized bulks from Figure 1, where the surface structures and Miller indices are shown, and considered adsorption sites are indicated with red dashed circles. Neighboring surface metal sites are either empty, OH, or O-covered depending on the configuration and unit cell size, generally corresponding to a high coverage regime. Explicit coverages are given in Figure 2. The surface configurations were chosen to span a large range of surface oxidation states to increase data set diversity. For MO_2 rutile, we selected two different Miller indices (110) and (100), and for M_2O_5, we considered fully covered as well as 50% covered surfaces. A 50% coverage was chosen for the M_2O_3 surfaces to include the +2.83 surface oxidation state. We only consider the most common OER type on-top surface adsorption, where the dangling metal bonds are occupied by oxo or oxyl-type oxygen bonded to single metal sites. Less OER active but often more stable bridge or peroxo-type configurations are excluded from the current study. Surface adsorption energies and the corresponding descriptors were consistently calculated with DFT and carefully converged in terms of atomic and magnetic structure, Hubbard-*U*, and pseudopotentials, as described in the Method section.

The adsorption energy results are summarized in Figure 2, with the left and right columns showing the adsorption of O and OH on 3d, 4d, and 5d oxides, respectively. Separate curves are used for each unique oxidation state and facet type. As an overall trend, the adsorption energy tends to increase (corresponding to decreasing bond strength) with increasing

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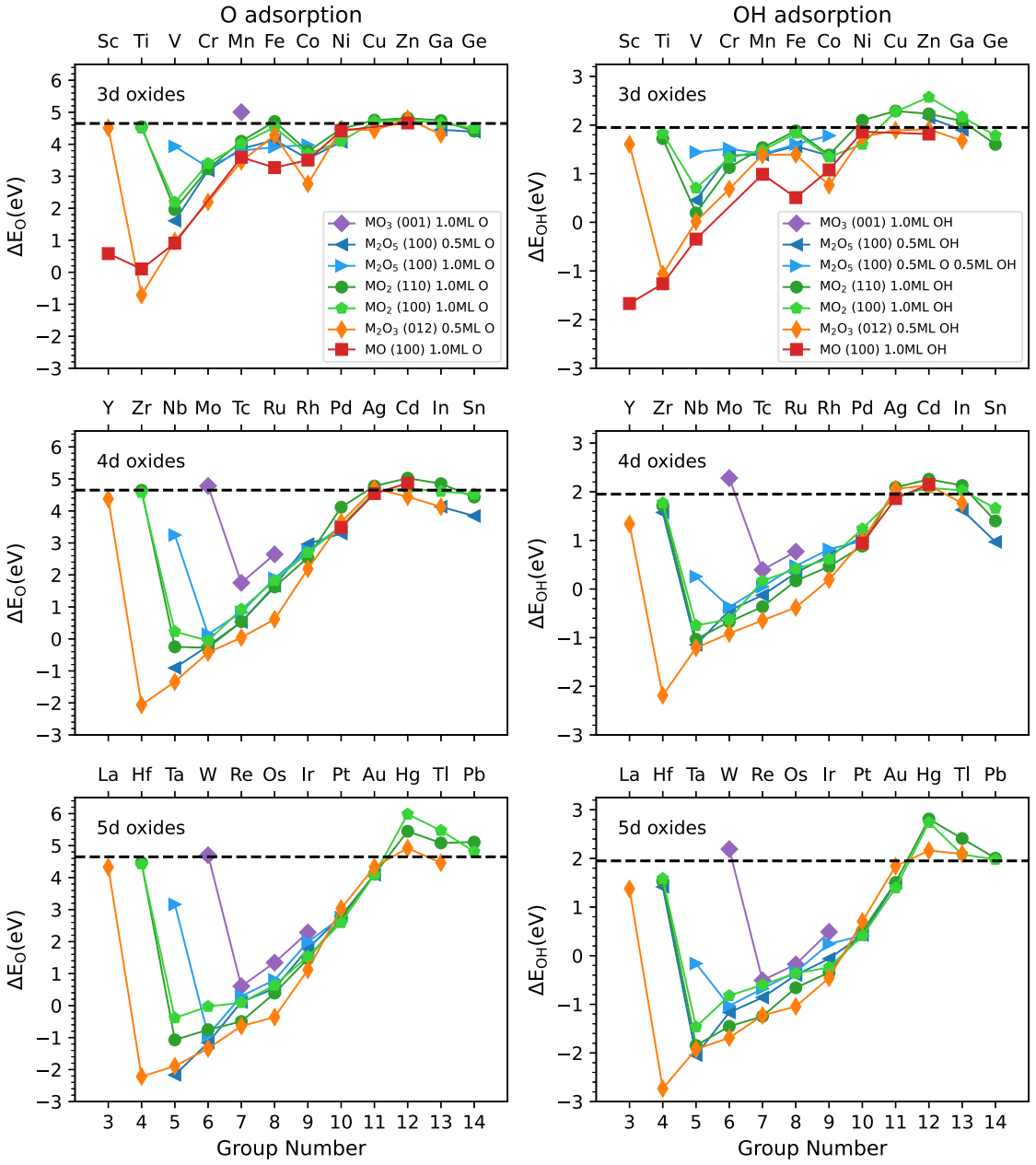


Figure 2. Calculated *O and *OH adsorption energies across the transition metal series for the +2, +3, +4, +5, and +6 oxide structures introduced in Figure 1. The legend indicates the systems and used coverage after adsorption, which applies to all sub figures in the same column. For the fully covered (1.0 ML) surfaces, the left and right columns are fully *O and *OH covered respectively, with exception of the +5 surfaces where the neighboring site is always O covered. The horizontal dashed line indicates the weak-binding limit for d^0 and d^{10} systems. The data set is available from www.catalysis-hub.org/publications/ComerGeneralized2024.

oxidation state for a given metal element; i.e., +3 generally binds stronger than +4, and +4 binds stronger than +5, and +6 binds the weakest from the whole series. This trend is expected from increased M−O coordination at higher oxidation states, leading to an oversaturation of the transition metal and increased repulsion between ligands which leads to weaker binding to the adsorbate. While relatively straightforward, this property directly controls the OER-relevant bonding strength in these systems.

Within each unique facet/oxidation state, we observe a *V*-shaped behavior starting initially with weak bonding followed by prominent discontinuity to strong bonding. This is followed by a gradual increase in the adsorption energy as a function of group number. Indeed, the realization of the *V*-shape adsorption is now observed by multiple researchers across multiple systems[20,21] which further underlines the importance of this observation. The weak bonding sites are due to the d^0 electronic configuration discussed in more detail below. The subsequent near-linear increase follows the trend previously reported by Calle-Vallejo et al. of increasing number of outer electrons,[22] as well as the trend shown for metals as explained by the d-band model.[13] In this linear part, as we move from left to right in the periodic table, the number of electrons increases and more antibonding states are filled leading to a weakening of the metal−oxygen bond. In our previous work we showed that this effect can be conveniently quantified from an ICOHP analysis of the M−O bond.[19]

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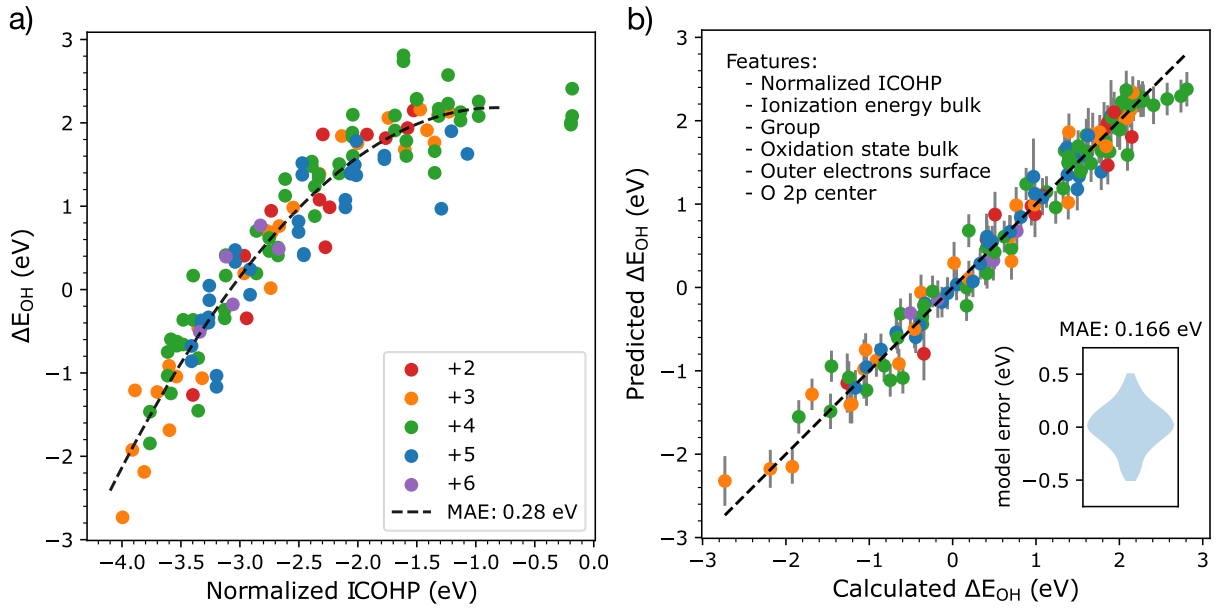


Figure 3. Model performance for the OH adsorption energy ICOHP-based models. (a) The OH adsorption energy plotted against the oxidation state normalized ICOHP with a 0.28 eV MAE deviation from the quadratic relationship discovered in previous work.[19] (b) Multidescriptor GPR ML model combining the ICOHP with bulk and non-DFT surface descriptors. Each point represents the leave-one-out cross validated model prediction, and error bars correspond to the standard deviation of the GPR predictive distribution. The analogous plots for O adsorption energies are shown in Supporting Information Figures S5 and S9.

The 3d metal oxides have more complex variations in adsorption energy which is not fully explained by the antibonding analysis. This complexity is a result of the high spin states on V, Cr, Mn, and Fe and Co^{2+}/Ni^{2+}, leading to more localized electrons and therefore less bonding with the neighboring oxygen atoms. For +3, +4, and +5 oxides, the spin leads to a discontinuity in the adsorption energy curve from Fe (high spin) to Co (low spin) both forming its own *V*-shaped sections. Interestingly, the size of the discontinuity decreases with increasing oxidation state of the transition metal in agreement with the prediction that increasing crystal field mitigates this effect.

Prominent discontinuities in the adsorption energies are also observed among the early transition metal oxides where weak adsorption is observed when the transition metal donates all of its valence electrons, i.e., reaches a d^0 configuration. The onset of this weak binding limit (shown as dashed horizontal lines) depends on the oxidation state of the metal, and therefore shifts to occur at group 3 for +3 oxides, group 4 for +4 oxides, etc. The d^0 transition also depends on the surface oxidation state and coverage as observed for the +5 V_2O_5, Nb_2O_5, and Ta_2O_5 surfaces where the d^0 transition is only observed for the high coverage configuration. For the 0.5 coverage configuration, on the other hand, strong adsorption is achieved by donation of charge from the unoccupied to the occupied site. Given our results, we focus our effort on developing models to predict the adsorption energetics in the non-d^0 systems. Specifically, we show how a generalized model, excluding the d^0-series, can be obtained by combining our bulk ICOHP with the additional bulk and non-DFT surface features of the oxide.

Bulk ICOHP as a Descriptor for Surface Adsorption. Crystal orbital Hamiltonian population is a method of analyzing and quantifying the nature and strength of bonding interactions within chemical systems.[23,24] The method is based upon partitioning of the band energy into interactions between projected local atomic orbitals. The COHP is calculated by multiplying the overlap between two atomic orbital projections by their associated Hamiltonian matrix element. When integrated up to the Fermi level, yielding the ICOHP value, an approximate bond energy is obtained as[25]

$$\mathrm{ICOHP} = \int_0^{E_F} \sum_j \langle \phi_{i,l}|H|\phi_{i',l'}\rangle\langle \phi_{i',l'}|\psi_j\rangle\langle \psi_j|\phi_{i,l}\rangle \tag{1}$$

where ψ_j are the band functions and $\phi_{i,l}$ are localized atomic basis functions with atom index i and angular momentum number l, where only the interatomic (off-diagonal) terms $i \neq i'$ are included. When based on DFT in the projector augmented wave method, the electronic states must first be transformed from the plane-wave basis set to atom-localized orbitals, where the projection is performed.

The ICOHP expression is closely related to the well-known Newns-Anderson model,[26−28] where the coupling between an adsorbate state a and surface state k, known as the coupling matrix element V_{ak}, is defined as

$$V_{ak} = \langle \phi_a|H|\phi_k\rangle \tag{2}$$

where ϕ_a and ϕ_k are localized (or tight binding) orbitals of the adsorbate and surface, and H is the Hamiltonian of their interaction. The off-diagonal elements determine the extent of coupling between the surface and adsorbate orbitals, which translates into the splitting of the metal's d-states into bonding and antibonding interactions.

Even though these two concepts are inequivalent, they both represent the extent of coupling between two atomic orbital states. As we show later, these states are of high relevance for representing the bonding strength between the surface and adsorbate. A benefit of our approach is that we can utilize states obtained from ab initio calculations of the bulk oxide rather than surface states to eliminate the need for surface DFT calculations. In previous work,[18] all atomic orbitals were included in the ICOHP of the metal oxygen bond, whereas in

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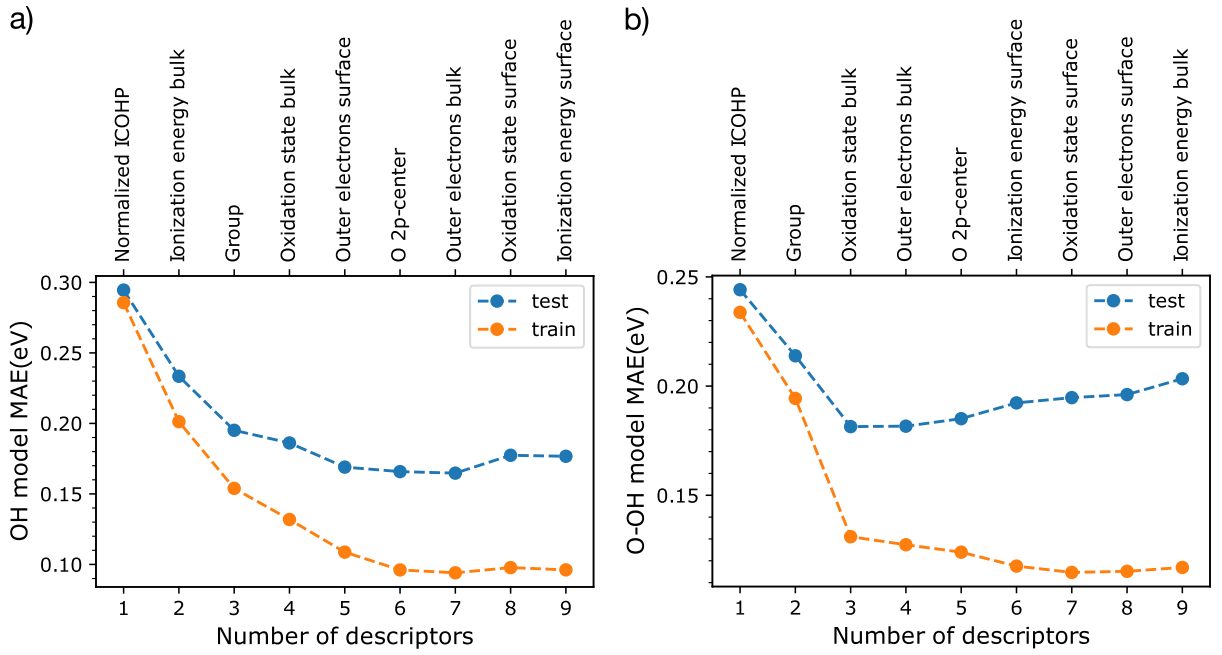


Figure 4. Result of the forward sequential feature selection algorithm for (a) OH adsorption and (b) O−OH adsorption energies. The train and test set errors are computed from averaging over leave-one-out cross-validated errors over the entire data set of 140 values.

our most recent work, we showed that the interactions between the oxygen 2p and metal d orbitals[19] have a particularly strong correlation with adsorption energies.

In this work, we expand on our previous approach to describe transition metal oxides across different crystal structures and oxidation states. By doing this, we find that to capture the variation across metal oxidation states, it is more favorable to include the entire s + d valence shell of the metal, so that both M(d)−O(2p) and M(s)−O(2p) interactions are included in the ICOHP sum, which is particularly important when comparing +2 to other oxidation states. Next, we address a diversity of crystal structures by including all nearest neighbor oxygen of the metal center. Specifically, for each bulk material/catalyst, we calculate the ICOHP of all bonds of a selected metal center to generate the ICOHP sum for a given M−O octahedron including the 6 nearest oxygen neighbors.

Furthermore, we can generalize across oxidation states by normalizing with the formal oxidation state of the metal, resulting in the following equation for the summed and normalized ICOHP over a metal center M

$$\mathrm{ICOHP}_{\mathrm{norm}} = \frac{1}{M(\mathrm{ox})} \sum_{i=1}^{6} \mathrm{ICOHP}_{\mathrm{M-O}_i} \tag{3}$$

where the sum is over the ICOHP of the 6 bonds of the metal center and $M(ox)$ is the formal oxidation state of the metal. In the Supporting Information Figures S1 and S2, we show that the non-normalized ICOHP increases with oxidation state, while the normalized ICOHP decreases in magnitude with oxidation state in good agreement with the change in adsorption energy. The normalization is needed because the (formal) number of electrons involved in the surface−adsorbate bond stays approximately constant, while the number of electrons involved in the M-O bonding in the bulk changes significantly with the oxidation state.

In Figure 3, we show the correlation between the normalized bulk ICOHP and the DFT calculated OH adsorption energies. We note that the case of d^0 systems is treated separately in the Supporting Information, and refer to Figures S3 and S4 for an in-depth discussion. Furthermore, the analogous plots for O adsorption energies of O and the adsorption energy differences of O−OH are shown in Supporting Information Figures S5 and S6). We see that the ICOHP successfully captures the trend in adsorption energies, recovering the quadratic relationship also seen in our previous work for the rutile series.[19] In the strong-binding regime, as expected, a large (negative) absolute value of the ICOHP corresponds to strong surface adsorption with a close-to-linear relationship. In the weak binding limit, however, the adsorption energy approaches a constant as the ICOHP goes toward zero. Further linearization can be achieved by considering the ICOHP differences at the surface level, which we will address in a follow up work. Although the normalized ICOHP qualitatively captures the trends in adsorption with a relatively low MAE of 0.28 eV for the quadratic fit, it is desirable to further decrease the prediction error of this simple method. In the following, we therefore study how the inclusion of additional descriptors within a ML framework can help further improve the model performance.

Multidescriptor ML Model. To improve the generalizability of the ICOHP-base prediction, we train a Gaussian process regression (GPR) model, where we consider additional descriptors that do not require any surface DFT calculations. Following approaches taken by other research groups[15,22,29,30] we have considered the features:

- metal group number.
- Formal oxidation state.
- Number of outer electrons, as determined by the metal element and oxidation state.
- The bulk oxygen 2p center (DFT).[14]
- The bulk normalized ICOHP as defined in eq 3 (DFT).
- The normalized ionization energies of the metal as defined in eq 4.

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For non-DFT quantities, we include both the bulk and surface values of a descriptor in our final feature set, resulting in 9 features in total. When evaluating the surface oxidation state, assuming a fixed oxygen charge of −2, we average over the formal oxidation state of all of the surface atoms (values shown in 1). For the ionization energy descriptor, we select the i-th ionization energy corresponding to the $M(\mathrm{ox}) \rightarrow M(\mathrm{ox}+1) + e^-$ transition (reflecting the ability of the surface site to be oxidized further), and normalize with the oxidation state

$$I_{\mathrm{norm}} = \frac{I(M(\mathrm{ox})+1)}{M(\mathrm{ox})} \tag{4}$$

Particularly, the normalized ionization energy as well as the O 2p center show a high correlation with the adsorption energy, as shown in Figures S5 and S6. However, for all adsorption energy targets (OH, O, and O−OH) the normalized ICOHP is superior as a stand-alone descriptor. The full correlation matrix between the adsorption energy targets and all considered descriptors is shown in Figure S7.

A forward sequential feature selection algorithm was applied to choose the most relevant features for the GPR model by adding one feature at a time and selecting the model with the lowest MAE. Due to the relatively small size of the data set (approximately 140 points for each adsorbate), we used leave-one-out cross validation for each point and averaged over all models to obtain the final model error. In Figure 3b, the performance for the best 6-descriptor OH adsorption energy model is shown. We obtain a low MAE of 0.166 eV and an RMSE of 0.213 eV, thus showing good performance and generalizability across all oxidation states.

Features are listed in order of importance in Figure 4a. We find that the normalized ICOHP gives the best 1-descriptor model, while the best 2-descriptor model is obtained by including the bulk ionization energy and so forth. By examining the evolution of the test and train set model errors, a 6 descriptor model was found to give the best model performance while minimizing overfitting. We applied the same model and feature selection approach to our O adsorption energy target as well as the O−OH difference, which is descriptive of the OER activity. We obtained MAE values of 0.219 and 0.181 eV for O and O−OH, respectively. All results are provided in the Supporting Information Figures S8−S10. Interestingly, we find that for the O−OH model shown in Figure 4b, only 3 descriptors are needed to obtain reasonable absolute errors: the normalized ICOHP, group number, and bulk oxidation state. Thus, the further addition of surface descriptors does not improve the model and its MAE, but leads to overfitting. This finding indicates that the O−OH energy difference is less dependent on the surface facet and coverage than the separate O and OH adsorption energies.

In general, we have found that the normalized ICOHP from the bulk DFT calculations both qualitatively and quantitatively captures the trends in adsorption energetics across oxidation states with better performance than any other single descriptors tested. Furthermore, when adding other non-DFT based descriptors, the generalizability and predictive accuracy of the model are further improved. We note that since the correlation between the ICOHP and O 2p center is high, the O 2p center is found to have low importance in the multidescriptor model. If the normalized ICOHP is excluded from the model, the O 2p center is found to be the best single descriptor, although with a reduced model performance.

In conclusion, both the ICOHP and the O 2p center of the bulk oxide are strongly correlated with the adsorption energies of the O and the ICOHP and are favorably included as ML features to improve the performance of adsorption energy models. However, the ICOHP shows the best performance as a single descriptor. As we discuss in the next section, an additional benefit of the ICOHP descriptor compared to the O 2p center is the close relationship between the ICOHP and the M−O bond length in the bulk oxides that can be utilized for an even less costly feature generation.

Toward Bond Length-Based Models. It is know that a strong correlation exist between the bond order and bond length.[31,32] Pauling[33] showed that the bond order of C−C bonds was related to the bond length through the simple relationship known as the Pauling form

$$r_n = r_1 - a\log(n) \tag{5}$$

where r_n is the bond length for bond order n, and a is a fitted constant. Similarly, the ICOHP (approximate bond energy) has been shown to correlate strongly with both bond length and bond order.[34,35] We thus transform eq 5 to relate ICOHP and bond length

$$\mathrm{ICOHP} = -\exp\left(\frac{-(r-r_0)}{a}\right) \tag{6}$$

where r_0 and a are fitted constants for each element. The a parameter controls the rate of decay of the ICOHP with increasing bond length and r_0 corresponds to the bond length at ICOHP = −1 eV.

We investigated the correlations within our bulk oxide data set by calculating the ICOHP for each bond around the metal centers. Figure 5 shows the fitted correlation between the ICOHP and bond length sampling the available bonds in all of the investigated oxidation states for a select set of metals. For some metal oxide structures (in this case the M_2O_3 and M_2O_5 polymorphs), there is a large variation of the M−O bond lengths, resulting in several distinct points for the same

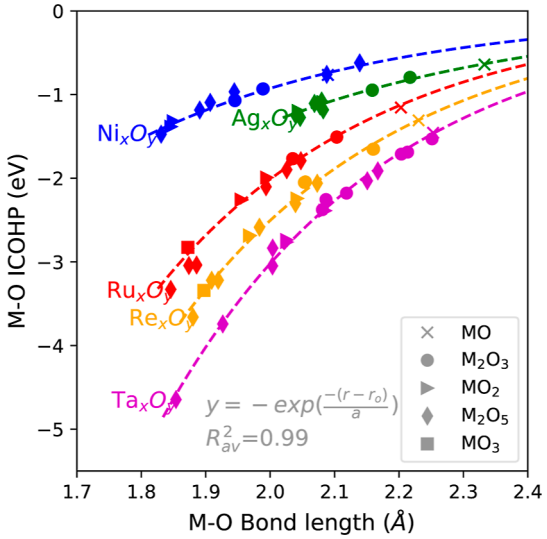


Figure 5. Correlation between bond length and ICOHP for the studied bulk materials. The functional form of the equation can be seen in eq 6. The fits are highly accurate, typically having R^2 of above 0.95. The full table of fits including R^2 values can be seen in the Supporting Information Table S1.

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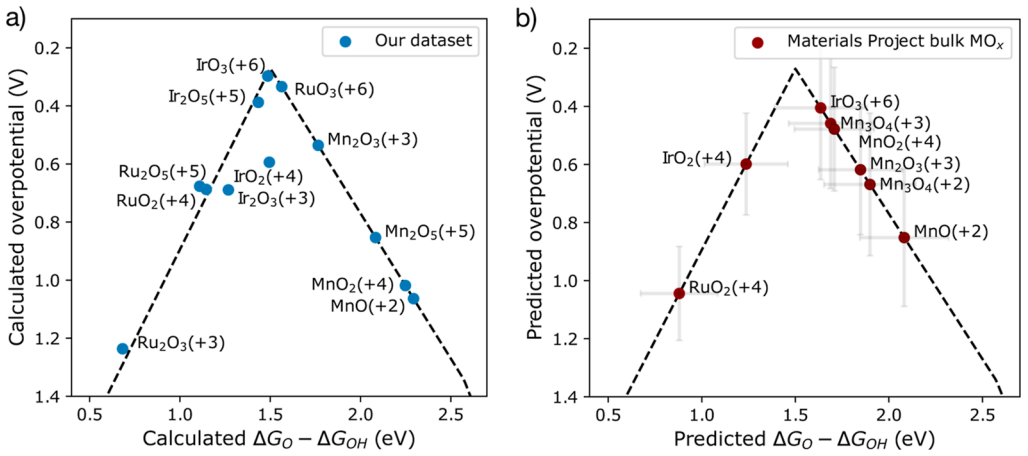


Figure 6. OER activity volcanos showing the (a) calculated overpotential obtained from O and OH adsorption energies shown in Figure 2 and (b) predicted overpotential for stable E_{hull} < 0.05 eV binary oxide structure sourced from Materials Project. The prediction is based on our O−OH 3-descriptor model, where the normalized ICOHP is obtained from the M−O bond lengths of the optimized crystal structures as given by eq 7. Error bars correspond to O−OH GPR model uncertainties for the *x*-axis, which is converted to *y*-axis (overpotential) error bars consistent with volcano line slopes.

structure. We also note that this variation is well captured by the individual ICOHP changes between the bonds. As expected, we find that the higher the oxidation state, the stronger (i.e., the shorter) the bond of the bulk oxide. We note that this trend is reverse to that of the normalized ICOHP that decreases in magnitude with oxidation state, consistent with the changes in adsorption energies.

The generated fits are very accurate, with R^2 values of 0.98 for much of the data set, where the relationship spans multiple structures and oxidation states. These correlations provide a direct connection between the metal−oxygen bond length and a quantitative measure of the bond strength in pure metal oxides. The fitted constants and corresponding R^2 values are tabulated in the Supporting Information Table S1.

As the ICOHP approaches zero, the bond length increases, in correspondence with the expected behavior of an interatomic potential. However, we note that the correlation only holds for optimized structures. For example, the ICOHP values of unrelaxed bulk bonds (with net interatomic forces) are not expected to fall on the curves shown in Figure 5, and the ICOHP will not sufficiently capture the repulsive potential between two atoms at low separations, where the true interatomic potential should diverge. In general, the true bond energy differs from the ICOHP, where additional terms, including the repulsive lattice energy and onsite COHP energy, and other terms contribute to the total cohesive energy of a system.[23]

Predicting Catalytic Activity Directly from Bulk Structure. In the last section, we focus on the direct prediction of the catalytic activity for the OER, where the theoretical overpotential can be obtained based on adsorption energies from Figure 2 as described in the Method section. For simplicity, we focus on a few high-activity metal oxides, selecting Ru, Ir, and Mn-based oxides, choosing the (110) surface facet series for rutile MO_2 and the high coverage series for the M_2O_5 surface. Coverage configurations are given in the legends of Figure 2. As shown in Figure 6a, the calculated overpotential follows a volcano relationship with the $\Delta G_O - \Delta G_{OH}$ free energy difference.

As expected, Ru, Ir, and Mn[36]-based oxides are found to be near the top of the volcano (low OER overpotential and high catalytic activity). An increase in oxidation state generally leads to a larger $\Delta G_O - \Delta G_{OH}$ value consistent with a weakened bond strength. From the calculated adsorption energies, we predict that higher oxidation states of Ir (+5, +6) will have a higher catalytic activity for OER, in agreement with previous work.[37−39] In the case of Mn oxides, however, the evolution in catalytic activity with oxidation state is nontrivial due to spin effects as mentioned above. We note that the activity of RuO_2 is generally underestimated from DFT calculations (placed too far left on the volcano), which has been attributed to a different OER mechanism[40] as well as problems with representing the magnetic structure.[41]

Next, we utilize our 3-descriptor GPR model for O−OH to estimate the OER activity for bulk systems. Given the strong relationship between ICOHP and bond length, we can approximate the ICOHP directly from bond lengths of the optimized bulk structure, i.e., without explicitly calculating ICOHP. Written in terms of the fitted constants ($r_{0,m}$ and a_m) for each metal, the normalized ICOHP value then becomes

$$\mathrm{ICOHP}_{\mathrm{bond}} = \frac{-1}{M(\mathrm{ox})} \sum_{n=1}^{N_o} \exp\left(\frac{-(r_n - r_{o,m})}{a_m}\right) \quad (7)$$

where $M(\mathrm{ox})$ and N_o are the oxidation state and the number of oxygen neighbors, respectively. As seen in Figure S11 the correlation between the ICOHP from bond length and calculated values is very precise with an R^2 = 0.99.

As a proof of concept, we now apply our proposed models to binary oxides in the materials project (MP) database.[42] The model includes (1) normalized ICOHP, (2) group number, and (3) bulk oxidation state as descriptors for the O−OH adsorption energy. First, we train the model on our calculated data set, using the DFT-based ICOHP as a descriptor. Next, we apply the model to MP structures by estimating the normalized ICOHP from the bond lengths, as described in a previous section. We note that since our bulk oxides have been calculated with a very similar calculational DFT setup and code, the bond length versus ICOHP correlations are comparable between these two data sets. Furthermore, we are only considering pure binary(MO_x) oxides with a E_{hull} < 0.05 eV. Since some of the oxides have mixed oxidation states

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for the same metal element, we are obtaining model predictions for each unique metal center, which also allows for rapid screening of different OER active sites.

The predicted O−OH energy is then translated into an OER overpotential through the O−OH descriptor and mapped onto the OER volcano. This approach can also be extended to the prediction of the ORR activity with the OH adsorption energy as the main activity descriptor. We note that since the O−OH GPR model is based solely on bulk features, no assumptions are made on surface facet cut or overages for the MP activity predictions. However, we expect the model results to best represent stable surface configurations with an oxidation state close to that of the bulk, generally corresponding to the high coverage regime given our training data. In Figure 6b, we show the predicted overpotential for Ir, Ru, and Mn-based binary oxides from the MP. The positions of these oxides on the OER volcano are found to be qualitatively well captured by our model, demonstrating the ability to predict the catalytic activity from bulk structure.

At first glance, the ability to predict surface properties and catalytic activity from the bulk structure might seem surprising. However, we are essentially utilizing the strong correlation between the bulk metal−oxygen and surface−oxygen bond strengths in systems outside the stable d^0 cases, where the metal element and oxidation state are found to be the key features. Another interesting finding is the negligible importance of surface descriptors in our multidescriptor model. In the feature selection algorithm, the surface features (oxidation state and ionization energy) were found to be less important than their bulk counterparts. Furthermore, for our O−OH energy difference model, the addition of surface features did not improve model performance and only led to further overfitting. This reflects that (within our data set) the O−OH difference is less surface facet-dependent than the individual adsorption energies. However, our model is unlikely to be directly transferable to unstable or metastable surface terminations or step edges of the same bulk crystal structure. This can also be seen as a strength of our model since the predicted activity will correspond to that of a more bulk-like, most stable surface configuration, even though that surface configuration is unknown. More work is needed to understand the relationship between the ICOHP and the adsorption in less stable sites or adsorbate types. Furthermore, more work is needed to understand the effect of bulk crystal structure (polymorph) variations and alloying within the same chemical composition. For example, COHP-based models will need to be extended to systems with large structural changes, i.e., from doping in mixed metal oxides[20] or with large nonlocal electronic effects, i.e., subsurface alloys, supports[21] where the O 2p plays an important role.

We envision that our approach can be particularly useful to accelerate the initial screening steps of catalyst material candidates and rapidly reduce the number of expensive surface DFT simulations and/or experiments. Subsequent computational steps should ideally involve more sophisticated computational methods to accurately capture the electrochemical environment of the surface, including the simulation of all relevant OER intermediates (H_2O, OH, O, and OOH) and explicit (or implicit) treatment of the solvent. In terms of the chosen DFT methodology, it is straightforward to extend our ICOHP approach to other DFT functionals or other choices of Hubbard-*U* corrections. However, we emphasize the need for experimental and future chemical accuracy benchmarks for adsorption on transition metal oxide surfaces.

CONCLUSIONS

In this work, we utilized a consistent data set of adsorption of O and OH on pure transition metal oxide surfaces to develop models for adsorption across oxidation states from +2 to +6. We showed that the metal−oxygen bond strength in the bulk oxide, obtained as the M(d + s)−O(2p) ICOHPs, strongly correlates with O and OH adsorption energies. When normalizing the ICOHP with metal oxidation state, we found this correlation to be universal across oxidation states, where a second order polynomial fit obtains MAE of 0.28 and 0.39 eV for OH and O, respectively, superior to other single descriptors considered in this work, including the O 2p center. Furthermore, we showed that when using the ICOHP descriptor in a simple GPR ML model, only a few additional descriptors are needed to reach MAE of 0.166 eV and 0.219 for OH and O adsorption, respectively, and we also found the O−OH adsorption energy difference to be well captures with a MAE of 0.181. Furthermore, we utilized the strong correlation between bond lengths an ICOHP to develop bond-length-based models for adsorption energies as well as OER activity. This opens a path toward predicting the OER/ORR activity of materials based directly from optimized bulk-only calculations.

METHODS

DFT Calculations. The DFT calculations presented in this work were done in the Vienna Ab initio Software Package (VASP)[43−48] at the PBE + *U* level of theory. Hubbard-*U* were used in the 3d metal systems as summarized in the Supporting Information Table S3. The exact values used for each metal can be seen in Supporting Information, Table S2. A planewave cutoff of 500 eV was selected and a *k*-point density of at least 25 Å was used for surface and bulk systems. All systems were relaxed to have forces below 0.02 eV/Å. Collinear spin polarization was used and the magnetic moments were manually inspected to ensure spin variations between the clean surfaces and the adsorbed surfaces was only present in the region around the active site. This methodology is consistent with our previous work.[19] For the +2 (rocksalt) and +3 (corundum) polymorphs, antiferromagnetic ordering was chosen, while ferromagnetic ordering was chosen for the rest of the polymorphs.

For the bulk systems, we considered all polymorphs and oxidation states for each metal and performed an additional nonself consistent DFT calculations to be used for the O 2p center and COHP calculations. These calculations were done with a fixed charge density (ICHARG = 11) with symmetry turned off (ISYM = 0) and a finer *k*-point grid with a density of at least 40 Å.

Electronic Structure Descriptors. The LOBSTER package was used to perform COHP calculations.[17,23,25,49,50] Calculations were carried out with the pbeVaspFit2015 basis set with the recommended VASP pseudopotential setup and including all metal oxygen bonds for a selected metal center. An energy grid from −25 to 15 eV was used. In the subsequent processing, only the metal valence s + d orbitals and the oxygen 2p orbitals are included in the sum to obtain the ICOHP value.

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The oxygen 2p center was obtained as the band center of the oxygen 2p projected density of states using a fixed integration range of −10 to 3 eV.

GPR Model. We applied a GPR model constructed from two squared exponential kernels with different isotropic length scale hyperparameters as well as a white kernel. Kernel length scale and white kernel (noise) hyperparameters are optimized during model training with a L-BFGS-B algorithm. Features are standardized to zero mean and unit variance before model training. For train-test splitting, we used leave-one-out cross validation and averaged the MAEs across N models, where N is the size of the data set. Model uncertainties (error bars) are reported as the standard deviation of the GPR predictive distribution. To select the most important features, we employed a forward sequential feature selection algorithm.

Calculated OER Overpotential. We assume a 4-electron mechanism through the OOH intermediate with the following electrochemical steps:

1. $H_2O + * \rightarrow OH^* + (H^+ + e^-)$
2. $OH^* \rightarrow O^* + (H^+ + e^-)$
3. $O^* + H_2O \rightarrow OOH^* + (H^+ + e^-)$
4. $OOH^* \rightarrow * + O_2(g) + (H^+ + e^-)$

The OER overpotential is then defined as the difference between the maximum free energy step (1−4) and the theoretical water splitting potential

$$\mu_{OER} = \max[\Delta G_1, \Delta G_2, \Delta G_3, \Delta G_4]/e - 1.23\ V \quad (8)$$

The OOH adsorption energy is approximated from the scaling relation with OH, and adapted as $\Delta G_{OOH} = 0.84\Delta G_{OH} + 3.14$ eV from previous work,[51] corresponding to OH coverages listed in Figure 2. Free energy corrections as tabulated in Table S3 are applied to DFT O and OH adsorption energies to obtain Gibbs free adsorption energies.

The volcano relationship between the calculated overpotential and the $\Delta G_O - \Delta G_{OH}$ is constructed from the O−OH (Figure S11) and OOH−OH scaling relations. In the case of our O−OH ML model, the O−OH GPR prediction values are then translated directly into an OER overpotential following the volcano relationship. Model uncertainties (error bars) are reported as the standard deviation of the GPR O−OH model predictive distribution for the volcano plot x-axis, while the error bars for the overpotential y-axis are obtained by scaling the uncertainties of the O−OH model with the volcano line slopes.

■ ASSOCIATED CONTENT

Data Availability Statement

The adsorption energy data set including surface slabs, gas phase, and bulk geometries are made available at www.catalysis-hub.org/publications/ComerGeneralized2024.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acscatal.4c00111.

Additional analysis for the non-normalized vs normalized ICOHP across the entire bulk series, as well as a more in-depth discussion about the d^0-oxides and how surface ICOHP calculations can be applied to understand these systems; adsorption energy parity plots between all adsorbates (O, OH, and O−OH differences) and the best performing single descriptors; feature correlation matrix, as well as feature selection results and model performance for all adsorption energy targets; fitting constants between the ICOHP and M−O bond lengths, as well as the calculated and predicted OER volcano plots for all the materials in the data set and MP for overpotentials <1.4 V; and additional computation details, including values for Hubbard-U corrections, free energy corrections as well as the O−OH scaling plot (PDF)

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Notes

The authors declare no competing financial interest.

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NOTE ADDED AFTER ASAP PUBLICATION

This paper was originally published ASAP on March 25, 2024, with an error in eq 7. The corrected version was reposted on March 26, 2024.

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